



Aim

Discuss the scope of winemaking and the set of characteristics

INTRODUCTION

The first thing to understand about winemaking is the difference between viticulture and oenology.

Viticulture is the study of growing grapes. Those grapes might be used for wine, but they might also be used as fresh fruit to produce juice (non-alcoholic) or dried fruit. Viticulture is concerned with growing the plant and producing the fruit.

Oenology is the study of wine and wine science. It is only concerned with the alcoholic product produced from harvested grapes. The production of fruit is only of concern in so much as how it affects the alcoholic beverage it is turned into.

GLOBAL WINE PRODUCTION

The global production of wine in 2018 was 292 million hectolitres. The major wine-producing countries (Italy, France, and Spain) recorded an approximately 30% increase in wine production.

World wine production in 2019 in million hectolitres (where 1 hectolitre = 100 litres):

- Italy 54.8
- France 48.6
- Spain 44.4
- USA 23.9
- Argentina 14.5
- Australia 12.9
- Germany 10.3
- South Africa 9.5
- China 9.1
- Portugal 6.1
- Russia 5.5
- Romania 5.1
- Hungary 3.6
- Brazil 3.1
- New Zealand 3
- Rest of the World 29.2



GLOBAL WINE CONSUMPTION

Global wine consumption was 246 mhl in 2018. The USA leads the world in wine consumption, consuming about 33.0 mhl, followed by France (26.8 mhl), Italy (22.4 mhl), Germany (20.0 mhl), and China (17.6 mhl). Australia (6.0 mhl) is the 10th largest consumer in the world.

WHAT IS INVOLVED IN WINEMAKING?

Winemaking is an accomplished art, and the quality of the wine depends greatly on the skill of the winemaker. Many winemakers start as enthusiasts and turn a passion into a living. Good winemakers are always in demand, and great winemakers are highly sought after.

The basic processes of most winemaking can be briefly summarised as follows:

- Removal of unessential material such as stems and leaves
- Grape crushing
- Fermentation
- Storage and Aging, or bottling

The basic difference between red and white wines is that in the production of white wines, the grape skins are removed prior to fermentation, whereas in the production of red wines, the skins are fermented along with the juice. Of course, whilst not nearly as popular, you can make wine from other fruits like elderberries or rosehips and use other things like rice. Grapes are ideal for winemaking because they are the only fruits which contain tannins, esters and acids which enable makers to produce stable and consistent wines.

Components of grape juice important for wine production are:

- Sugars – glucose, fructose
- Organic acids – tartaric acid, malic acid, citric acid
- Tannins – catechol, chlorogenic acid, caffeic acid
- Amino acids and proteins
- Minerals – phosphates, sulphates
- Vitamins – b-group, ascorbic acid
- Important volatile components that affect flavour and colour

WINEMAKING TERMINOLOGY

Must

This is the fresh-pressed grape juice used to make wine. It is the raw material to which yeast is added and left to ferment.

Tannins

Tannins are a type of polyphenol found in plants. Tannins are found in many types of fruit, including the skin, stem, and seeds of wine grapes. Tannins are also found in wooden barrels, sometimes used to age wine. Most tannic wines are dry and somewhat astringent. Tannins bring a certain complexity and bitterness to wine; wine that is “too tannic” or “overly tannic” is strong and bitter.

The quantity of tannins in wine depends on how long the stems, seeds, and skins are soaked in the juice. Longer soaking times mean higher tannin content, which is why red wines contain more tannins than their white counterparts.

Tannins are also antioxidants, which protect wine from oxidation and changes in flavour caused by exposure to air. It should be noted that the taste of tannins softens with age, reducing the “harshness” of the tannic taste over time.

Phenols

These are a diverse group of chemical compounds that affect taste, colour (especially anthocyanins), and even mouthfeel in wine. Like tannins, phenol concentrations can be increased by letting the juice “macerate” or sit with the grape skins. Detailed management of phenols may not be necessary for small-scale winemaking but can be of concern for commercial and high-quality wine production.

Sugar



Sugar is an essential part of winemaking. It drives fermentation, contributes to the taste of the wine, and has an effect how well it ages and is preserved. Winemakers use several terms referring to sugar. The percentage of sugar in the must is the simplest of these measures. Two other measures are also common:

- Degrees Brix (or simply, °Brix) - is essentially the same as 'degrees balling' (or °Balling) which is also used in some scenarios. This is a measure of the percentage of sugar at a particular temperature measured using a hydrometer.
- Degrees Baumé - which is approximately half the value of °Brix.

It is also possible to estimate the sugar content of wine via the specific gravity (also known as relative density).

Acidity and De-acidification

Must or wine with a high acid content can be de-acidified with commercially available products. The acidity of the must or wine can be measured by pH. Acidity affects taste. Low levels of acidity result in flat-tasting wine. If levels are too high, the wine may taste too sharp.

Sulphites

Sulphites are added to wines to stop further fermentation, oxidation and spoilage.

Sulphur dioxide

Sulphur dioxide is an anti-microbial agent used as a preservative for musts.

Extract

Wine extract is the solids in wine. It includes sugar-free extract, minerals, and non-volatiles. Extract provides a wine's body and colour. Water, sugar, and alcohol do not form part of the extract.

Filtration Processes

Filtration refers to the removal of undesirable particles from wine. Various types of filters may be used to achieve this.

Racking

Racking is part of the filtration process. Wine is pumped or siphoned off the lees (or sediment) into another vessel when it is racked. Wine should be racked soon after fermentation because if it remains in contact with the lees for too long, it may develop bad flavours. A typical wine would be racked between two and four times. The removal of the lees ensures greater clarity.

Wines finished after primary fermentation are usually racked quickly. Wines set aside to undergo malolactic fermentation are racked once or twice but not fully because the lees meet the nutritional requirements of the malolactic fermentation bacteria.

The timing of racking depends on the wine's specific gravity and the density of the lees. In some cases, racking may be required during malolactic fermentation since this type of fermentation may take several months. Over this time, a large quantity of lees may produce bad flavours.

After all fermentation is finished, the wine is racked once more. It should be noted that whilst racking is beneficial, there is a limit to how often it should be done because each episode allows some oxygen into the system, which can accelerate the wine's aging.

Fining

A wine that has yet to be racked has a cloudy appearance. The presence of suspended protein particles causes this cloudiness. Fining agents are used to get rid of these particles. They are materials that cause proteins to bind together and drop to the bottom of the wine. In doing so, racking and filtration become a lot easier, which means that fewer racking episodes are needed. Removing these proteins quickly also reduces the likelihood of any bad flavours developing in the wine. Two of the most common fining agents are bentonite and gelatine. There are others available.

Filtering

The final step in finishing a wine before bottling or ageing is filtration. This is done to remove any fine particles. The type of filter used depends on whether or not the wine has been sweetened after fermentation. A particle filter is usually used for unsweetened wines. This type of filter will remove any remaining suspended particles. If the wine has been sweetened after fermentation, or has any residual sugar, a sterile filter must be used. These are especially fine filters that remove almost all microbes that may be present in the wine (due to its sugar content).

Anything else that comes into contact with the wine must also be sterile so as to maintain a microbe-free state. This includes bottles and corks. The wine must also be transferred straight from filter to bottle to reduce exposure to air. Filtering wine may cause it to taste flat, but this flatness will disappear after several weeks.

TESTING, TASTING, MONITORING

Converting grapes into wine can be exceedingly complicated, and the finished product (the wine) can vary significantly in taste, aroma, and visual appearance.

- Aroma refers to what is perceived by smelling the wine
- Taste refers to what is sensed by tasting the wine.
- Visual appearance considers the clarity and colour of the liquid.

As the process of making wine progresses, the liquid goes through various changes in its chemical and microbiological characteristics, which need to be monitored. Interventions are made along the way to control what is happening, and ultimately the wine which is produced should be relatively stable so that it can be stored until it is drunk, without deterioration in the overall quality.

Discrimination Testing

A technique used in sensory analysis to assess if two or more items have a measurable difference. The test uses a group of evaluators (panellists) with a degree of training suitable for the difficulty of the test to distinguish between products through a range of experimental designs, such as:

- Triangle test: A useful multipurpose test where a taster is required to select different samples. This test uses 3 samples two that are identical, and one is different. It has good application in determining if process change affects the overall product characteristic.
- Duo-Trio test: This test is sometimes used instead of a triangle test to compare wines unknown difference. Tasters are presented with a reference wine and then two test wines. Evaluators are asked to identify the sample that is same as the reference wine.
- Paired comparison test: It can be used where there is a known difference in the chemical composition of wine. Also, to see if there is a directional difference for a single characteristic.
- Same/Different comparison test: Similar to paired comparison test, however it is used when the difference between two wines is unknown.

Preference Testing

Preference testing refers to consumer tests in which the consumer is given a choice to look at one product directly against a second product, as in product improvement studies or when determining parity with a competitive brand. This technique also works very well when the consumer panellists have minimal reading.

For wine studies, consumer evaluations from various perspectives have also been conducted such as studies considering consumer subgroups, clusters, or demographic data, or time-related studies. Many factors must be considered due to the complexity of wine, with the involvement of sensory attributes including intrinsic/extrinsic factors and others such as target consumers and methods.

WHAT CAN GO WRONG IN WINEMAKING

It is essential that people who assess the quality of wine can accurately recognise the common sensory wine faults and taints that occur in wines. Wine faults are defined as off-aromas or tastes related to the fermentation or the winemaking process. Wine taints are defined as an odour, taste or aftertaste that originates from external or foreign sources. Some of the common wine faults are as follows:

- Oxidation: Caused when wine is exposed to too much oxygen. Wines that are oxidized will give off aromas that range from dull to cardboard and straw, and stewed fruits.
- Acetaldehyde: Associated with oxidation of ethanol into acetaldehyde. Addition of SO₂ during fermentation can increase the concentration of acetaldehyde, as can increases in pH and fermentation temperature.
- Volatile acidity: Occurs when bacteria create acetic acid. Often associated in tanks or barrels with excessive headspace, low SO₂ and poor winery hygiene. Affected wines smell of vinegar or acetone, like that of nail polish remover.
- Hydrogen sulphide (H₂S) : Reduction (by yeast) of elemental sulphur, amino acids etc. into hydrogen sulphide (H₂S). It can result in rotten egg, pungent, sulphur like smell; easily detected and dissipates quickly with aeration.

- Dimethyl sulphide: Major compound found in aged wines and formed during wine maturation. The aroma is described like canned corn, cooked cabbage; garlic, burnt rubber.
- Sulphur Dioxide: Excessive addition particularly at bottling. It has sharp, pungent smell.
- Trichloroanisole or Cork Taint: Caused by moulds growing on cork coming into contact with chemicals used in disinfectants. It causes the wine to take on a dank musty and mouldy odour that smells like wet newspaper or cardboard. Avoid bleach, chlorinated solvents and high levels of Cl- in water.
- Geraniol: Normally produced by lactic acid bacteria from sorbic acid. Distinctive geranium odour, floral, sweet, pungent smell. Maintain proper levels of SO₂
- 4-Ethyl phenol: Characteristic aroma from Brettanomyces (spoilage yeast). Proper sanitation, maintaining sulphur levels, and eliminating problem barrels is essential. Sterile bottling recommended for problem wines. This compound imparts woody, band aid, medicinal character to wine.
- 4-Ethyl guaiacol: Characteristic aroma from Brettanomyces (spoilage yeast). Wine is described as having Warm, spicy, medicinal, band-aid, pungent, clove/spice aroma.

ALCOHOL AND HEALTH

When discussing alcohol and alcohol consumption, it is impossible to do so without also talking about the health impacts of alcohol. Generally, the study of alcohol and health focuses on the volume of alcohol consumed by individuals and whether they are daily consumers or binge drinkers. As already outlined, statistics relating to consumption may be skewed by errors. Statistics relating to personal alcohol consumption are also prone to errors because most people underestimate their alcohol consumption or are not particularly forthcoming due to social stigma.

Binge drinking hurts health. Whilst different authorities have slightly different explanations for what constitutes binge drinking, one definition that seems to be quite widely accepted is that binge drinking is where a person drinks a lot of alcohol in a short space of time so that their blood alcohol level is 0.08g per decilitre. This usually occurs when a woman consumes four drinks or a man consumes 5 drinks in less than 2 hours. In the United States about 90% of alcoholic drinks consumed by people aged 21 years and younger are said to be through binge drinking. Most people who binge drink do so on an average of 4 occasions per month, and one in six people in the US binge drink.

Binge drinking is associated with a higher prevalence of violent and aggressive behaviour, increased likelihood of hurting oneself and others through accidents or violence, and an increased risk for suicide. When it comes to alcohol dependence, Russia has the highest rate with as many as 20 percent of adult males being dependent. Europe and Latin America have high prevalence rates with typically three to four times as many men affected than women. Latvia and Lithuania have the highest recorded rates for alcohol-related hospitalisations at 2,00 per 100,000 residents.

One of the reasons why alcohol-related health problems have increased in some countries is due to a lack of suitable policies to control the marketing and advertising of alcohol. A lack of education concerning the health consequences and cultures that actively encourage the consumption of alcohol also contribute. In countries where alcohol-related problems and the amount of alcohol consumed per capita have decreased, this is often linked to better policies and awareness. The implementation of drink-driving laws, restrictions on advertising, increased taxes on alcohol and greater availability of alcohol-related treatment services have all been effective in this regard. The World Health Organisation has implemented a Global Information System on Alcohol and Health (GISAH) as a kind of watchdog that presents data and information on alcohol consumption and alcohol-related problems along with the effectiveness of policies to control its consumption. It features data from more than 100 countries.

We believe that anyone who makes wine, beer, spirits, or other alcoholic beverages should be aware of the health risks to themselves and others and take a responsible approach to the production and consumption of alcohol. With this in mind, making alcohol can be a fun and satisfying pastime or career.